# Toward an Integrative Theory of Higher Education:
## Connecting Lines of Inquiry from Morin's Complex Thought, Bhaskar's Critical Realism, and Wilber's Integral Theory

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To attempt to provide an adequate reading of the interrelationships between Edgar Morin's Complex Thought (CT), Roy Bhaskar's Critical Realism (CR), and Ken Wilber's Integral Theory (IT) feels like something of a daunting task. To attempt to then draw out implications for higher education adds a further dimension of challenge. To do all of this in the short space of an article seems a mission impossible. And yet – from what perspective is such an impossibility imagined? It would presumably include the perceived necessity of sufficient detail and precision; indeed, academic writing and the analytical mind is predicated on this requirement. But if allowance is made for layered expression (perhaps analogous to layered ontology and layered epistemology), then it might be that we could move from the necessity of the binary of *either-precision-or-vagueness* toward the allowance of a more nuanced articulation of being *clear* (at one level of expression) about when certain types or degrees of *vagueness* might be appropriate (at another level). This paper is exploratory, initiatory, indicatory: preparatory work (to be coupled with other preparations) toward an integrative theory of higher education. A little fuzzy logic therefore feels apt in trying to form a short singular narrative pointing to the panorama of horizons offered by these three meta-approaches.[1] We should surely owe our allegiance here to the scholarship of integration (Boyer, 1990) and its prospective interest in meshing, metaphor, and meaning. A certain degree of clustering will therefore be enacted, one which minimizes contestabilities between the three meta-approaches in favour of highlighting similarities.

The paper first outlines the three meta-approaches – both in themselves and with respect to (higher) education.[2] This is then followed by a more in-depth indication of their interconnections. A particular limitation is that whilst much of Morin's work is in French and Spanish, the paper is bounded by Anglophone research. This limitation is ameliorated both by English-language versions of Morin's key works such as *Method* (1992) and *Seven Complex Lessons in Education for the Future* (1999) as well as by useful English-language overviews of Morin such as those given by Montuori (2004, 2008).

### Morin's Complex Thought

Morin was born in Paris in 1921, making him more than two decades older that the founders of the other two meta-approaches, and the only one whose work has been directly influenced by the Second World War (Morin, 1946; Montuori, 2008). Morin, a sociologist by training, was originally an active member of the French Communist Party, but was subsequently expelled in 1951. At the time of his retirement, Morin was Emeritus Director of the National Center of Scientific Research and co-founder and Director of the prestigious Edgar Morin Centre of the School for Advanced Studies in the Social Sciences, both in Paris. He is presently involved in the creation of the Multiversidad Mundo Real Edgar Morin[3] in Mexico. His writing has a "continental" flavour, in some ways reminiscent of his contemporary French philosophers such as Derrida (Montuori, 2004): it tends to be somewhat poetic, playful, humble and self-reflexive. He also has an ongoing propensity to collaborate creatively with artists – something rather rare in the social sciences (Montuori, 2004). His work uniquely integrates such discourses as complex systems science, poststructuralism and neo-Marxism

into an encyclopaedic, transformative whole – one imbued with both humour and curiosity. Morin's (1992, 2008) writing has been driven by the search for a method that would enable understanding of the complexity of humanity. This quest forms the basis for the development of earth-citizenship, and concomitant ethics for the "human genre" (Morin, 1992, 1999, 2004, 2008). It is facilitated through taking action against "knowledge-mutilating" principles – such as disjunction, closed specialization, undue reduction, and "false rationality" – in favour of pertinent knowledge (Morin 1999). Morin also adopts a dialogic principle in his work; this entails bringing ideas (both complimentary and contradictory) into relationship with one another in order to better identify and understand the complex nature of reality (Morin, 1992; Dobuzinskis, 2004; Montuori, 2004). Morin's dialogic emphasis stems from his Hegelian roots (Montuori 2004, 2008; Doduzinskis, 2004). It differs from Hegel's dialectic, however, in that it does not offer any guaranteed synthetic resolution (Montuori, 2004). All the above can be identified as CT – an approach not without its challenges (Morin, 2006)[4]. Ontological features of CT include the foregrounding of open systems, feedback loops and a holographic geometry. Phenomenological reality is understood as emerging and evolving in complex systems through a process of recursive organisation – "self-eco-re-organizing systems." Epistemological features of CT stem from an understanding of the organisation of knowledge and its history – leading Morin to be oriented by transdisciplinarity. Through this, the global, multidimensional, complex and contextual are addressed; there is an integration regarding the role of the knower; and its meta-paradigmatic character enables the questioning of disciplinary assumptions. Transdisciplinary is identified by Morin as mostly inquiry-driven, and a degree of creative eclecticism is allowed. Human beings are identified as *homo complexus* (with all the dualities that entails), whilst due recognition is given to the difference between signifiers and signifieds, coupled with the resultant care required when attempting to speak about ontology.

## Complex Thought and (higher) education

Morin is the only one of the founders of the three meta-approaches that has written extensively on education. The most thorough discussion occurs in the UNESCO-sponsored book, *Seven Complex Lessons in Education for the Future* (Morin, 1999). Its key points can be summarised as follows.

- *Lucid veracity in education* ("Detecting error and illusion") – Morin points out that we humans have psychological and cultural propensities which make us vulnerable to error and illusion. Apropos, part of education's job should be to directly address such topics as rationality and rational uncertainty (including due self-vigilance). The situation is made worse by adherence to "blinding paradigms" such as the Cartesian dissociation of objective and subjective knowledge. Furthermore, ideas should be understood as mediating, but distinct from, reality: the only ideas we should trust are ideas that include the idea that the real resists the idea. Education should therefore facilitate the competence of construct-aware lucidity.
- *Pertinence in education* ("principles of pertinent knowledge") – Morin argues that the dominant thrust of education is toward the disjuncture of closed specialization, reductionism, and "false rationality": knowledge becomes dislocated and petrified. Such a template obscures attempts at aptly addressing the global, the multidimensional, the complex, and the contextual. "We should teach methods of grasping mutual relations and reciprocal influences between parts and the whole in a complex world" (p.1).
- *Complex human identity in education* ("teaching the human condition") – Morin notes that human beings possess – at the least – physical, biological, cultural, social, and historical dimensions: we are *homo complexus*. Education should therefore directly

address our complex nature. The disciplinary structure of knowledge, on the other hand, disintegrates – even "mutilates" – this complexity. Transdisciplinarities are therefore called for (Morin 2008).

- *Planetary consciousness in education* ("Earth identity") – Morin argues that education needs to prioritise addressing the planetary nature of our present and future situation – we all share the same fate. Such a perspective would address both the beneficial and destructive effects of the rapidly accelerating geo-political inter-connectivity of the Earth since the 16th Century. This includes address of the contemporary ecological crisis.
- *Uncertainty in education* ("confronting uncertainty") – Morin indicates that whilst modern science has revealed many certainties, it has also provided us with at least a similar amount of uncertainties – in the physical, biological and social sciences. Given this, he argues that these uncertainties should play an essential part of education. There should be a prioritisation of the teaching of complex systems and strategic principles for aptly addressing chance, chaos, and uncertainty.
- *Understanding in education* ("understanding each other") – Morin indicates that a key purpose of education should be the development of understanding rather than the mere acquisition of information. He identifies communication as operating both inter-objectively and inter-subjectively, and that understanding is both the means and ends of communication. As a first step, education is required to identify and overcome obstacles to understanding, including egocentrism, ethnocentrism, and reductionism. He states that, "if we learn to understand before condemning, we will be on the way to humanizing human relations" (p.52), and that such ethics of understanding are developed through comprehensive thinking, continuous introspection, and open heartedness.
- *Ethics in education* ("ethics for the human genre") – Last but not least, Morin addresses the interrelationship between human individual, society and species with respect to critical awareness and action ("anthropo-ethics"). In this, he sees education as playing a core role. He posits that "moral lessons" are insufficient for engendering the ethical sense required for true human expression and that instead it is necessary to cultivate a deep understanding of what it means to deeply embody individual autonomy, participate in community, and be aware of the responsibilities inherent in belonging to the human species. This ethical awareness helps to prepare us for the move towards the ethical-political imperative of realizing our earth citizenship by way of democracy and fulfillment of humanity as a planetary community.

There is some reference made to Morin in discourse addressing complexity approaches to education (e.g. Mennin, 2010), but these should be distinguished from literature which shows a more thorough integration of Morin's ideas. Although Morin draws inspiration from complex adaptive systems research, his work has a far richer character than a narrow interpretation of this field: the move is from complex systems to a complexity paradigm (Morin 1982). CT brings together a range of seemingly disparate perspectives united by a deep yet subtle spiritual impulse (Montuori, 2004). As a result he is able to overcome many of the pitfalls of other complexity approaches, such as those identified by Brown (2010). Much of the educational literature arising from CT addresses *critical* education (e.g. Alhadeff-Jones, 2010). This suggests more direct links to critical education than is found in the other two meta-approaches examined. Morin's approach has also had significant impact upon the practice of academic research, both from the perspectives of complexity (Alhadeff-Jones, 2009) and transdisciplinarity (Del Re, 2008; Del Mello, 2008). Morin's dialogic approach, along with his epistemological and ontological position, necessitates an explicitly transdisciplinary engagement with knowledge (see, e.g. Morin, 2008).

**Bhaskar's Critical Realism**

Roy Bhaskar first began to develop a new philosophy of science whilst working towards a Ph.D in Economics exploring the relevance of economic theory for under-developed countries at Oxford University. His approached developed over three decades of independent scholarship, and he now holds the position of 'World Scholar' at the Institute of Education, London University. The discursive point of departure for Bhaskar's work is therefore that of the philosophy of science, specifically in relation to the British analytical school (Hartwig, 2008). A key purpose has been to create a viable alternative to positivism, a paradigm which has dominated scientific discourse in the modern era (Bhaskar, 1978). In so doing, Bhaskar has established himself as one of the most significant realist thinkers in the last thirty five years (Baehr, 1990).[5] CR has continued to grow in influence[6]. It nonetheless remains on the periphery of philosophical discourse (Shipway 2002): its radicality is seen not to easily connect with any "epistemic point of view that has contemporary standing" (Corson 1991a, p.189). In terms of style and tone, Bhaskar's writing tends to be dense and critical; the prevalence of new terms and acronyms leads to a comparison with Hegel's writing (Hedlund-de Witt 2012). Ontologically, CR involves a transcendental realism – a realism which has ontological depth, one which hierarchically nests the domain of the empirical within that of the actual which itself is nested within the real (Bhaskar, 1978). Other key ontological features include the emergent character of ontological strata[7], and a clear differentiation between intransitive and transitive dimensions[8] (Bhaskar, 1978). Additionally, dialectical thought indicates the primacy of ontological absence (Shipway 2002; Bhaskar 2008). Bhaskar distinguishes between open and closed systems, and also introduces the notion of the laminated system in which it is impossible to refer to one level of the system without reference to others (see, e.g. Bhaskar *et al*., 2010). One example of such a laminated system is the four planar social being[9]. Bhaskar argues for an emergent or evolutionary model of reality by recognizing seven evolving levels of human agency and collectivity – seven scalar social being[10] (Bhaskar in Bhaskar *et al.* 2010) – and seven stadia of the ontological-axiological chain of being[11] (Bhaskar 2002a; 2008; 2012). Epistemologically, CR views science as a social activity about real things: a process in which transitive knowledge is generated in reference to intransitive mechanisms (Bhaskar, 1978). Additionally, Bhaskar posits that truth may be conceptualized as comprising concepts of *referential detachment*[12] and *alethic truth*,[13] as well as a concept which he refers to as the *judgment form*[14] (Bhaskar, 2008). The attempt to reduce ontology to epistemology is identified as the *epistemic fallacy* (Bhaskar 1978; Collier 1994). Other concerns include the attempt to reduce the real to the actual (*actualism*) and the attempt to account for ontology solely with regard to the positive, rather than acknowledging absence (*ontological monovalence)* (Bhaskar, 2008). Methodologically, the notion of *explanatory critique* problematizes the imaginary that facts can be absolutely differentiated from values; whereas *immanent critique* is a philosophical method which reveals internal contradictions within a philosophical or sociological system by employing its own logic[15]. Alongside these critical approaches, the notions of induction and deduction are transcended by *retroduction*, "the move from a manifest phenomenon to an idea of a generative mechanism, which if it were real would account for the phenomenon in question" (Hedlund-de Witt 2012, p. 9). In terms of the disciplinary orientation of epistemology, CR valorizes inter- and trans-disciplinarity.

**Critical Realism and (higher) education**

Although Bhaskar has not written much on the field of education explicitly, he is presently editing a book series on critical realist education.[16] Of particular note in the CR literature on higher education is Ronald Barnett's (2012) *Imagining the University*. Barnett uses CR to

dismantle prevalent conceptions of the university, and instead proposes that we imagine feasible utopias of the university built upon imaginative ideas and an optimistic, sensitive understanding of the deep structures underlying the institutions of higher learning. As a philosophy of science, CR has direct implications for research – see, e.g. Scott (2005) regarding the relevance of CR to empirical educational research.[17] Corson (1990a, 1990b, 1991a, 1991b, 1997, 1997, 1998, 1999a, 1999b) provides a wide-ranging critique of particular philosophical approaches in education (most notably Dewey, Popper, and Quine). Additionally, Archer (1979) has provided important foundational work in the application of CR to the macroscopic investigation of how educational systems come into being. CR is most frequently encountered as an alternative knowledge paradigm being used as the basis for educational inquiry. It is used as the ontological (transcendental realist), epistemological (critical relativist), and methodological (explanatory critique) approach for specific research inquiries in the field of education (McGray, 2012; Soetaert, 2008; Loveday, 2009; Shipway, 2002).[18] It is interesting to note that of the four theses identified, two deal with the organizational aspect of higher education (in the broadest possible sense): Loveday (2009) explores cross-functional collaboration at community colleges, while Soetaert (2008) offers a CR-based analysis of strategic change in higher education. Shipway's (2002) thesis[19] stands out as the only doctoral thesis (to our knowledge) that deals exclusively with the implications of CR for education in a more holistic sense. Shipway expands upon the role of CR as an antidote to many of the problems facing educational research. He also provides an in-depth and wide-ranging exploration of the implications of CR for curriculum development as a test case for the application of CR in the philosophy of education. Shipway also examines the relevance of CR's emancipatory mission to the field of education. This is a theme which has subsequently been picked up by McGray (2012) whose doctoral research explores a CR approach the topics of capacity building and citizenship education in light of global hegemony. The implication of the emancipatory and ethical aspects of CR are thus far reaching, indicating the emergent social-change agenda with broad implications in the educational field.[20] The value of CR for the development of curriculum is further expanded by Wheelahan's (2012) sociological arguments for the central value of academic knowledge in curricula. Additional applications in the literature along these lines include critical realist engagement with instructional design and technology (Evans, 2011), assessment (Dobson & Dale, In Press), the emancipatory capacity of science education (Zembylas, 2006), emancipation with respect to learning disabilities (Warner 1993) and the education of indigenous populations (Sarra, 2012).

**Wilber's Integral Theory**

Wilber is an autodidactic American philosopher with a background in transpersonal psychology and Buddhism. He is sometimes announced as one of the most popular, and most translated, of contemporary philosophers. His writing often has something of an extraverted style. Whilst his writings prior to 1997[21] can be understood in relation to creating and contributing to IT, IT as such emerged explicitly in Wilber (1997). This emergence was enabled by his magnum opus, *Sex, Ecology, Spiritualty* (1995). Evolution is identified with respect to Eros, a universal or "kosmic" driver, whilst Eros is located with respect to the contrasting forces of Agape (descend-and-include), Phobos (ascend-and-exclude), and Thanatos (descend-and-exclude) (Wilber 1995). Wilber also forwards Koestler's notion of *holon*, a concept which indicates that all things are simultaneously wholes and parts. Additionally, Wilber brings together the insights of other integral theorists including Sri Aurobindo and Jean Gebser. Regarding ontology and epistemology, Wilber argues that they are "two mutually enactive and co-existing dimensions interactively resonating in the living

Kosmos" (Wilber, 2002). Modernist reductionism is concomitantly critiqued as a "flatland" ontology. IT ontology (see Esbjörn-Hargens 2010) and epistemology (see Fuhs n.d.) are understood through his AQAL model (Wilber, 1996, 1997, 2000a, 2000b, 2007a).[22] Methodologically, IT indicates the legitimacy of an integral methodological pluralism which uses 1st and 3rd person (i.e. interior and exterior) views of each of the four quadrants to render eight distinct perspectival zones, each of which is resonant with a particular type of methodology – ranging from phenomenology to systems theory (Wilber, 2007a).

**Integral Theory and (higher) education**

Although Wilber has not written much on education, he has indicated that it is one of the most crucial areas of application for IT (Wilber, 2000, 2008). Following Wilber's explicit identification of IT in 1997, a few educational articles have mentioned Wilber in passing (Miller, 2001), but the focus on education (including higher education) with respect to IT has developed in a more focused way following Esbjörn-Hargens (2005). Two volumes on Integral Education were published in 2010 – Esbjörn-Hargens, Reams & Gunnlaugson (2010a) and Dea (2010a) – the former solely addressing higher education (in the broadest sense), including reference to non-Wilberian forms of integral, e.g. Ferrer, Romero & Albareda (2010), and Ryan (2010). Themes identified from the literature are exemplified by the following:

- *Contexts, history and theoretical identification* – Hampson (2010), Molz & Hampson (2010), Gidley (2010), Esbjörn-Hargens, Reams & Gunnlaugson (2010b), Murray (2009);
- *General educational considerations* – Esbjörn-Hargens (2005, 2010), Toroyson (2010), Steckler & Torbert (2010), Reams (2010a);
- *Curriculum design, course development, and institutional renewal* – Renert & Davis (2010), Kreisberg (2010), Stack (2010), O'Fallon (2010a), Wheal (2000), Lloyd & Wallace (2004), Zierer (2011), Schmidt (20102); specifically including *spirituality* – Iannone & Obenauf (1999), Astin (2010); and *ecological sustainability* – Conway (2012) Renert & Davis (2012), Akiyami, Li & Onuki (2012);
- *Pedagogy/andragogy*, including the significance of: *interior perspectives* – Martineau & Reams (2010); Gruber (2010); *dialogue* – Gunnlaugson (2010); Reams (2010b); Bronson & Gangadean (2010); and *experiential embodiment* – Wieler (2010); O'Fallon (2010b); Dea (2010b); and Dea (2010c);
- *Assessment* – Klein (2012), Davis (2010a; 2010b), Dawson & Stein (2011); and
- *Case studies* such as regarding healthcare education – Kreisberg (2010), Pesut (2012).

**Connecting lines of inquiry**

The following provides an indicative reading of the connections between the three meta-approaches with respect to higher education. Its lenses include normativity, andragogy, the philosophy of knowledge (implicating both research and curriculum), and address of the whole educational system.

*Normativity*.[23] The three meta-approaches offer an ecosystem of normativities regarding higher education. A reading of the broad landscape of ethics from a developmental perspective is offered by IT. This includes the notion of widening spheres of responsibility and care arising with greater developmental depth – for example, from ego-centric through nation-centric and world-centric to kosmo-centric. CR's notion of the individual's self-structuration of being (stadia of the axiological-ontological chain) and of the hierarchy of levels of scale (or span) at which social phenomena can be studied (the seven scalar social

being) offers an analogous, although not identical, picture. Apt concern for the health of all levels can be identified as facilitated by Basic Moral Intuition – "protecting and promoting the greatest depth for the greatest span" (Wilber, 2000b, p.510). Within such a framing, CT indicates the critical pertinence of addressing the *world-centric* level arising from humankind's unique individual↔society↔species feedback loop (Morin 1999), and its concomitant identification of human as planetary citizen. Such a prioritisation would have radical implications for such fields as neoclassical economics, which, through such imaginaries as *homo economicus*, is based on assumptions – and thence de facto encouragement (Morin 1999) – of ego-centric self-interest and national interest rather than planetary interest. Recent trends regarding the "pragmatisation" of course priorities toward such fields as business and technology can also be questioned with respect to such spheres of responsibility: how much are such moves on behalf of the planet as a whole?

Of particular pertinence for higher education is the aim of emancipation (Morin, 1999; Montuori, 2008; Alhadeff-Jones, 2010; Bhaskar, 2008, 2012) – both social and spiritual, and as interpreted through different developmental levels.[24] CT is associated with critical theory Montuori 2004; Alhadeff-Jones, 2010) and its interest in *social* emancipation. This is perhaps no surprise given Morin's early involvement in the ideals of communism. Bhaskar's socialist heritage – indeed, the "critical" of Critical Realism – also has a resonance with CT in this respect. In terms of the critical nature of the fate of humanity and many other species caused by human-made changes to the planet, Morin's work is very direct in its address, whilst CR provides a robust philosophy of science for such concerns. Additionally, vectors toward *spiritual* emancipation or Enlightenment are emphasized in CR's meta-Reality and in IT. Emancipation could be understood as resulting from the natural, evolutionary process of holonic emergence arising from the tensions of Eros↔Agape, and Agency↔Communion[25] (Wilber 2000a). This can be identified as a dialectical process involving the transformative manoeuvre of absenting absence. In the case of higher education, this can be interpreted as helping move stakeholders from unwanted/unneeded sources of determination to wanted/needed ones (Bhaskar, 1986; Collier, 1994; Shipway, 2002). The potential of emancipation as a key purpose of higher education has significance not only for such normativity to be explicitly integrated in social science, education and the humanities (perhaps in reference to the classical philosophical question pertaining to The Good) but also indicates the pertinence of highlighting the critical context of natural science and technology: the consequences of scientific discovery and technological application are not socially or ecologically innocent; neither are funding priorities. From a different angle, the various ways in which the three meta-approaches address postformal developmental levels[26] also imbue the higher education with an uncommon sense of directionality and purpose. In general, contra to a business-as-usual paradigm with respect to the purposes of higher education, the three meta-approaches surely beckon a radical reorientation, one sitting more in reference to such aims as human potential, social well-being and biospherical resilience (Hampson, 2011, 2012).

*Andragogy and heutagogy*.[27] Through the axiology of freedom, CR valorises, *inter alia*, equality of individual worth regardless of individual potential (Shipway, 2002). The collective extension of this can be understood as leading to CR's and CT's emphasis on democratic processes (Collier, 1994; Shipway, 2002). Such understandings have implications for the way in which higher education is conducted (e.g. inclusive decision making and ownership of the learning process). The democratic impulse of CT and CR synergises with CT's prioritisation of learning through dialogue. It similarly resonates with IT's address of the development of collective interiors through dialogue and developmentally appropriate communication (Gunnlaugson, 2010; Reams, 2010b; Bronson & Gangadean, 2010). The teacher's role is

consequently less one regarding the transmission of knowledge and more one regarding the nurturing of meaning, and the facilitation of understanding and emancipation away from illusory (less real) imaginaries. Concomitantly, learners are seen less as objectified knowledge receptacles and more as developing subjects with context-specific histories and trajectories. A particular andragogical strategy is the facilitation of particular states of consciousness – both in terms of embodying the dimensions encompassed by the AQAL conceptual map (e.g. Dea, 2010b; Wieler, 2010; O'Fallon, 2010b) as well as with respect to sensory and experiential engagement (Gruber, 2010; Dea, 2010c). The importance of engaging individual interiors through meditative practice (Martineau & Reams 2010) is also valorised. These andragogical sensibilities oriented by individual interior (e.g. contemplative) and collective interior (e.g. dialogical) emphases sit in contrast to conventional expectations in higher education with respect to the more technicist ("transmission of information") andragogies implied by mainstream course structures. The meta-approaches further indicate that such interior dimensions – as well as exterior ones – can be aptly understood with reference to postformal developmental levels. Cook-Greuter (2003), for example, describes how the integral nature of the emergence of the self-identify developmental line impacts upon dimensions of being, doing, and thinking. Such framings can substantively assist in foregrounding learner-centered (Schiro, 2007) and process-oriented (Kelly 2004) andragogies (and associated *currere*) in addition to those regarding "dialogical action" (Freire, 1997). Despite such possibilities, explicit incorporation into university courses both of positive adult development and forms of complex reasoning beyond (Piagetian) formal operations remain a neglected potential.

*Philosophy of knowledge*. A pertinent feature common to the three meta-approaches regarding approach to knowledge is interest in addressing what might roughly be called *the whole,* or more precisely to be signified by a semantic ecosystem[28] involving the terms unity, whole, system, organisation, cosmos, planet (CT); totalities, constellation, system, cosmic envelope, unitary ground state of meta-Reality, nonduality (CR); holarchy, tetra-arising, Kosmos, Being, Spirit, Isness, The One, The All (IT). This differs from the dominant *de facto* approach to knowledge in higher education which carries the iterative mark of modern atomism (Hampson 2010), as indicated, for instance, by the dominant prioritisation of disciplines and sub-disciplines over the interdisciplinary and transdisciplinary. The *differentiation* of this whole can be identified in its diverse complexity via concepts not only such as physical, biological, psychological, cultural and societal (CT), but also through discernments including mechanisms, events, experiences, domains, dimensions, strata (CR); and perspectives, holons, types, states, levels, lines, and quadrants (IT). Relationships among these aspects – and between these aspects and the whole – can be further elaborated by using such understandings as complexity, recursion, feedback loops (CT); dialectics, lamination (CR); intersubjectivity, interobjectivity, translation and transformation (IT). Particular emphases include the notion of higher education as assisting we humans in becoming increasingly conscious of the evolution of consciousness and its complex (e.g. non-linear) character (see, e.g. Midgley, 2002). Such meta-theoretical lenses frame the complex "nature of nature" (Morin, 1992) (including human nature) in ways which are often at variance with mainstream higher education approaches. Of specific note, the significance of transitive interiority (how things are relatively experienced on the "inside" by both individuals and groups/cultures) is highlighted by all three meta-approaches.[29]

Both the valorization of transitive interiority and that of frames addressing apt coherences of academic approaches (epistemologies, methodologies etc.) are enabled by the notion of transdisciplinarity[30] (a notion which has implications for both research and curriculum).[31] Explicitly mediating between the "two cultures" (Snow, 1998) of the humanities and natural

science, transdisciplinarity (in contrast to disciplinary atomisation) can more adequately address such complex issues as the human-induced ecological crisis (including global heating, radical biodiversity loss, systemic pollution, and resource depletion). It is also synergistic with the field of higher education, given that education is a particularly complex *open* system (Corson, 1998), one in which CR understandings regarding the causal efficacy of reasons can become a valuable part of sound educational research; and given transdisciplinarity's ability to aptly inquire into the character of particular educational traditions. A particular interpretation of transdisciplinarity is articulated by Nicolescu (2002). This involves relations between ontology and epistemology, multidimensionality, transculturalism, ethics, spirituality and creativity; whilst transdisciplinarity as the "mode 2 knowledge production" of Gibbons et al. (1994) is characterised by complexity, hybridity, non-linearity, reflexivity, and heterogeneity. From a slightly different angle, CT identifies transdisciplinarity's "core conditions" as including: its inquiry-driven orientation, a reflexivity regarding meta-paradigmatic assumptions, a sense of the whole regarding the organisation of knowledge, the prioritization of contextualization, and the explicit inclusion of the knower (Montuori 2008). Transdisciplinarity involves at least two levels: whilst at specific levels there is an understanding regarding theoretical and methodological *plurality*, at the meta-theoretical level there is a sense of *coherence* or unity. Together these combine as unity-through-complexity. The strong transdisciplinary orientation of the three approaches is concomitant with an interest in providing an appropriate balance regarding the status of the humanities vis-a-vis natural science, and the spectrum therebetween. Explicit in this interest is a serious contestability toward logical positivism and/or its overapplication. CR offers an alternative philosophy of science to its singular imaginary; IT aptly constrains it – or at least post-positivism – into one quadrant (honouring other modalities of legitimacy for the other three quadrants). Similarly, what might be termed a fundamentalist approach to relativism is also contested by all three approaches. Instead, relativism is relativised, and context-dependency is implicated as dependent on context: the transitive dimension of constructivism has its place (the left hand quadrants of psychological and cultural realities) but should not *unduly* interfere with other domains[32]. This has implications for the humanities and for poststructuralism.

*Spirituality*. In contrast to the default imaginaries of both secular modernity and traditional religiosity, an integrative theory of higher education oriented by the three meta-approaches would also have to take spirituality into account. The identification of spirituality is amply articulated through IT (see, e.g. Wilber 2007), and also through Bhaskar's (2012) address of meta-Reality. IT frames emergence in spiritual terms and locates mystical and meditative traditions as inquiring into and regenerating the perennial philosophy (Wilber 2000a). Whilst spirituality is less explicated in CT, Montuori (2004) nonetheless identifies a subtle spiritual impulse inherent in Morin's Complex worldview as underpinning the totality of his work. Whist spirituality is discussed within higher education discourse (e.g. Shahajahan, 2010; Rockafeller, 2006), its import is still largely underenacted – perhaps thwarted by disciplinary Balkanisation (constrained within the sub-discipline of Comparative Religion): the baby of trans-rational Spirit is still mostly thrown out with the bathwater of sub-rational mythic religion.

*Complex integration*. Last but not least, the integrative character of the three approaches indicate the desirability[33] of addressing higher education as a complex, integrative whole. What relations between higher education and society, politics, economics? What influence and holistic effect of bureaucracy and accountability culture? Such questions should be explicitly addressed in order to move forward to a sufficiently comprehensive integrative theory of higher education.

**Ends-in-view**

There is considerable wealth of insight potentially offered to higher education by Morin's Complex Thought, Bhaskar's Critical Realism, and Wilber's Integral Theory – even more so when the lines of inquiry arising from the three begin to be connected. Unpacking and regenerating the complex relationships between the three is quite involved, including the address of divergent points of departure, shared understandings, intra-contestabilities, and coherent ways forward. An integrative theory of higher education would, of course, *inter alia*, require "depth diving" into a myriad of specifics. Nonetheless, through seeding multiple points of departure, we hope that the paper has indicated its potential including some key aspects of the territory. The paper has connected inquiry lines regarding philosophy of knowledge (including complex integration, transdisciplinarity and spirituality), andragogy/heutagogy, and normativity (including ethics and purpose). Such a preliminary exploration begs for further research. This could no doubt take the form of analyses of differences between the three meta-approaches. Yet, due to the urgent contemporary context, a research emphasis on finding a *common* way forward – and taking positive action toward this in the lifeworld – would seem even more valuable.[34] This general manoeuvre would be supported by a nuanced and extended interpretation of the scholarship of integration (Boyer, 1990), its criticality enhanced by the recursive notion of *the pertinence of pertinence*. What would be necessary for more impactful uptake? The scholarship of integration counters undue (fundamentalised) modern atomism (Hampson, 2011, 2012). It needs to be sufficiently rigorous, complex and open. Applied to higher education, a resultant integrative theory could perhaps be more accurately understood as a postformally cohered ecosystem of higher education poetics (see Hampson, 2007, 2010). The following possible *research and action vectors* are indicative[35] of such a call.

*Contexts of integrative higher education theory*,[36] including:

- *Significations.* What terms should be used (and in which discursive contexts)? What reflexive explication should be given regarding relations between signifiers and signifieds? What degree and type of semantic clustering could be aptly employed in order to facilitate meaning and advancing together?
- *Sensibility*. What allowance of intra-contestabilities? Or the quality of the integrative theory as an open system? What security? What vitality? What power? What heart? What expression? What insight? What spirit?
- *Historical and cross-cultural contexts.* What macro-contexts would be useful to address? What learning for the field of higher education could be gleaned?
- *Contemporary lifeworld contexts*. What are the most pertinent critical contexts of the contemporary situation? What emphases of purpose (and concomitant process and structures) should be therefore empowered by higher education?
- *Contemporary discursive contexts*. What is the ecosystem of discourses which informs, or could inform, higher education? Given emphases of purpose arising from the above considerations, are there any discourses which could benefit from prioritising? What of higher education in futures studies? Or higher education as a form of social innovation? What types of conceptual insights for higher education arise – both actually and potentially – from the three meta-approaches (separately and jointly) discussed in this paper? What enables apt hermeneutic coherence? Additionally, what influential non-academic discourses are there? And what relations between the academic and non-academic? What key interests do each of these have?

*Content of integrative higher education theory*. Domains ripe for coherence include:

- *Ethics*. What would facilitate moves from narrow, modernist conceptions of ethics in higher education (e.g. research protocols and personnel issues) to broader, reconstructive postmodern (Griffin, 2002) conceptions regarding overall purpose? What ethical relevance derives from consideration of future scenarios for humanity, other species, and ecosystems?
- *Collegiality*. How might the university foster authentic interpersonal dialogue and development among and between faculty and students? Can the university of the future nurture and enact an adequate feeling of community – university as an *a*ffective community of scholars, staff, and students (see Hampson, 2010b)?
- *Andragogy/heutagogy*. What apt forms of learning occasions can be identified?
- *Architecture*. What aesthetic sensibility? What ecological impact? What relations to the "inner curriculum" of courses or the "hidden curriculum" of institutional culture?
- *Administration*. What relations to faculty? What is possible, desirable?
- *Pre-tertiary education*. What relationship to primary and secondary education? To what extent and in what ways do these categories foster fragmented silos?
- *Informal education*. What connections should there be between formal higher education and the learning that occurs in non-institutional environments – including that with peers (e.g. through social media)? How could the formal and informal be cohered? What role for technology?
- *Evaluation*. What would be apt forms of evaluation and assessment? How might certain types of evaluations form undesired consequences or feedback loops?
- *Philosophy of knowledge and action* – including (i) *Transdisciplinarity*. What would enable the divergent voices of the various disciplines to be sufficiently honoured whilst simultaneously maintaining (a possibly evolving) transdisciplinarity frame? (ii) *Spirituality*. What ways might contemporary spiritual directions (or higher reason) inform the purpose, fabric and process of higher education? (iii) *Reflexivity*. How can reflexivity be aptly enacted?[37] (iv) *Our organic substrate*. What is the significance of human biology to higher education? What apt connections can we identify between our organic nature and that of the planet?[38]

In general, we hope that seriousness is given to the quest to put "higher" (back) into higher education, and the *uni*-verse[39] (back) into the university – for the sake of life on this precious planet.

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[1] The Incompleteness Theorems of early twentieth century mathematician Kurt Gödel can be understood as indicating that (most) systems can either achieve self-consistency or comprehensiveness but not both. So perhaps such inclusion of "soft semantics" in the current instance are inevitable, even desirable.
[2] Whilst some of the literature explicitly refers to higher education, much addresses education in general (inclusive of higher education). We have included both.
[3] Among other courses, this university offers a range of qualifications (from Diploma to Ph.D level) addressing various aspects of the application of complex thought in the field of education.
[4] These challenges include: a realisation of the limits of conventional logic; irreducible disorder in complex relationship with organisational order; the intricacy of biological and social phenomena; the notion of clear distinctions (e.g. between an object and its environment, or between an observed phenomenon and the observer) as not necessarily offering a criterion of sound reasoning and methodological appropriateness; and the role of the observer in his or her own observation (Morin 2006).
[5] His work has passed through three phases (Hartwig, 2012; Hedlund-de Witt, 2012): (i) (basic) CR, which focuses on transcendental realism in the natural sciences, and critical naturalism in the social sciences (Bhaskar 1978, 1979, 1986); (ii) dialectical CR, which brings CR into a rich relationship with dialectics – notably the philosophies of Hegel and Marx (Bhaskar, [1993] 2008, 1994); and signals the beginning of Bhaskar's "spiritual turn"(Hartwig, 2012) with the introduction otTranscendental Dialectical CR (Bhaskar, 2000); and (iii) the Philosophy of MetaReality (Bhaskar 2002a, 2002b, 2002c, 2012), including the deepening of the aforesaid spiritual turn toward such notions as an infrastructural "absolute reality" or "ground state" (Bhaskar 2012). For convenience, our paper identifies all three phases under the umbrella term, *CR.* This should not be confused with other approaches that have identified themselves as Critical Realist including theological critical realism. A thorough discussion of the various uses of the term 'critical realism' can be found in Shipway (2002).
[6] The CR movement has become increasingly fragmented in recent years as CR's Marxist proponents have responded with distress to Bhaskar's metaphysical turn (Shipway 2002).
[7] CR argues that nature is stratified. This implies that there is a basic hierarchy of being with some ontological strata being more basic than others. This effectively describes the way in which being emerges through a hierarchical process of evolutionary complexity and is – in some ways – similar to the holarchical principals developed by Koestler (1967) and later developed by Wilber (1995). For example, the actions of living creatures are governed by the laws of the natural (inorganic world) – animals are constrained by the laws of physics and chemistry – however these laws are not sufficient to explain their workings; because living organisms are also subject to the laws of biological systems thus. Thus, anything belonging to a higher stratum is necessarily governed by more than one set of laws as – to borrow Wilber's (1995) terminology - each successive stratum transcends and includes the ones which precede it. We can thus observe a movement towards greater complexity moving, in broad strokes, from the physiosphere (the world of material beings), to the biosphere (the world of material beings), to the world of rational beings (Collier 1994).
[8] The intransitive refers to underlying mechanisms which form the ultimate objects of scientific inquiry, whilst the transitive refers to the relativistic, subjective and intersubjective aspects of science.
[9] Bhaskar (in Bhaskar *et al.* 2010: p.9) writes, "This specifies that every social event occurs in at least four dimensions, that of material transactions with nature; that of social interactions between humans; that of social structure proper; and that of the stratification of the embodied personality. These four planes constitute, of course, a necessarily laminated system of their own in so far as reference to any one level or dimension will also necessarily involve reference to the others".
[10] These are identified by Bhaskar (in Bhaskar et al. 2010: p.9) as: "(i) the sub-individual psychological level; (ii) the individual or biographical level; (iii) the micro-level studied, for example, by ethnomethodologists and others; (iv) the meso-level at which we are concerned with the relations between functional roles such as capitalist and worker or MP and citizen; (v) the macro-level orientated to the understanding of the functioning of whole societies or their regions, such as the Norwegian economy; (vi) the mega-level of the analysis of whole traditions and civilizations; and (vii) the planetary (or cosmological) level concerned with the planet (or cosmos) as a whole."
[11] The first four stadia (identified during Transcendental Dialectical Critical Realism) are: (i) First moment (1M) deals with 'being as such and as structured non-identity; (ii) Second edge (2E) deals with being as process or 'becoming'; (iii) Third level (3L) deals with being as totality; (iv) Fourth dimension (4D) deals with intentional transformative agency, transformative praxis; The last three (identified in the Philosophy of MetaReality) are: (v) Fifth aspect (5A) deals with being as incorporating reflexivity (and spirituality); (vi) The sixth realm (6R) deals with being as re-enchantment; (vii) The seventh zone (7R) deals with being as non-duality (or as involving essential unity) and therefore exists as the ground state of all being (7 Z/A)
[12] Referential detachment is "the ontological dislocation of referent from the act of reference" (Bhaskar 2008, p.131).

[13] Alethic truth refers to the truth of things which exists independently of propositions. Propositional truth presupposes alethic truth.
[14] The judgment form is the basis of judgmental rationality and is the formal method that can be used to distinguish between the veracity of competing theories. This form is built of the stratified nature of truth and involves examining theories in terms of expressive veracity, fiduciariness, descriptive veracity, and evidential proof (Bhaskar 2008, p. 221).
[15] The self-reflexive application of this critique undoubtedly contributes to the robustness of CR theory.
[16] Entitled *New Studies in Critical Realism and Education,* which is pulling together a number of the leading advocates for critical realism in the field.
[17] Scott argues that it is essential that empirical research is underpinned by a metatheory which gives it a clear epistemological and ontological orientation, one which acknowledges real objects of inquiry (which scientific exploration logically necessitates) whilst not committing one to the spurious belief that only one interpretation of knowledge is possible (or ultimate). CR is, in his opinion, the most appropriate metatheory to address these concerns. Furthermore, he redresses the unconstructive dualisms that exist between structure and agency in research. This has implications for research into the field of higher education as well as research as an educational activity.
[18] It is often referred to as the 'theoretical' approach or perspective (e.g. by McGray and Soetaert) or as a 'philosophical' approach or orientation (e.g. by Loveday).
[19] Unfortunately Shipway's research was completed ahead of the development of PMR.
[20] Extending beyond curriculum and organisational critiques to include andragogy, and foreseeably other aspects such as assessment.
[21] Wilber's work is often understood as being divided into five phases (Helfrich 2007): Phase I (1972-1978) – "recaptured goodness," or the Romantic Jungian (Wilber, 1977, 1979); Phase II (1978-1983) – "growth to goodness" (Wilber, 1980, 1981, 1983); Phase III (1983-1993) – holonic (Wilber, 1983, 1986, 1991); Phase IV (1993-2000) – holarchic, or AQAL (Wilber, 1995, 1996, 1997, 1999, 2000a, 2000b); and Phase V (2000-) – Integral Post-Metaphysics (Wilber, 2007a, 2007b).
[22] AQAL stands for *all quadrants, all levels, all lines, all states, and all types*. Quadrants are identified as four "tetra-arising" perspectives of reality rendered from two essential distinctions, namely, individual-collective and interior-exterior. Levels describe the evolutionary way in which development emerges in stages. Regarding human individuals, for example, these include postformal levels of complexity, indicating psychological development beyond Piagetian formal operations (including transpersonal psychology). The understanding is that each level offers a different type of true-but-partial view, and that more advanced levels "transcend and include" (in Hegelian term, *sublate*) preceding ones. Lines describe different types of developmental phenomena such as emotional development and cultural worldview (Esbjörn-Hargens, 2009), whilst states describe transitory shifts of consciousness. Gross states include waking (gross), dreaming (subtle), and deep sleep (causal) (Wilber, 2007a) whilst other states form in response to internal or external stimuli (e.g. hormonal states, altered states). Other examples might include weather states. Last but not least, types are identified as relatively durable structures or characteristic formations identified within each quadrant (e.g. personality types, governance types).
[23] In terms of the relationship between normativity and description, their intertwined nature is highlighted by Morin, and given an explicit concept – "explanatory critique" – by Bhaskar.
[24] IT does not tend to use the term *emancipation*. This is perhaps due to the fact that Wilber's thought finds its lineage in Hindu and Buddhist philosophy (the perennial philosophy) and contemporary transpersonal psychology, whereas Bhaskar an Morin share neo-Marxist roots.
[25] (tensions which are resolved in the non-dual heart)
[26] A large number of detailed adult development frameworks now exist which provide a detailed understanding of the unfolding of development across multiple lines, including: cognitive (e.g. Commons, Richards & Armon 1984; Kegan 1982; Kegan 1994); psycho-spiritual (Aurobindo 1975); psycho-social (Erikson 1978); moral (Kohlberg 1984; Gilligan 1993); values and worldview (Graves 2005; Beck & Cowan 1996; and Gebser 1991); and self-identity (Loveinger 1976; Cook-Greuter 2003); and perspectives on axiological-ontological depth (Bhaskar 2012) . [MERGE WITH:] Building on Loveinger (1976) and Cook-Greuter (2003) research into the development of self-identity, Fisher, Rooke & Torbert (2002; Rooke & Torbert 2005)- later followed by Joiner & Joseph (2007) – worked towards refining the articulation of action logics. These frameworks - especially when read in conjunction with, for instance, Kegan's work on 'orders of consciousness' - provide valuable insight into the mode of operation at different levels of development.
[27] Due to increasing levels of transformative agency and self-reflexivity associated with emerging ontological depth, a developmental understanding also foreshadows the movement from pedagogy to andragogy (Knowles 1984; Knowles 1990) to heutagogy (Hase & Kenyon 2000).

[28] Within which the terms interrelate complexly – involving intersections, parallels, divergences, and contestabilities.
[29] CT identifies psychological and cultural dimensions; CR clearly differentiates intransitive and transitive dimensions; whilst interior perspectives (both individual and collective) are central to Wilber's framework. Of corresponding note is a critique of monolithic positivist orientations, and uncohered relativist positions: CT identifies both of these in terms of the fragmentation and mutilation of knowledge (Morin, 1992, 1999); CR identifies them as classic sites of the epistemic and ontic fallacies (Bhaskar 1978); and IT addresses them as quadrant reductionisms (Wilber 2007a).
[30] CR emphasizes the term, *interdisciplinarity*. Whilst CR enables linking between disciplines, it more greatly offers an overarching philosophy of science or "underlaborouring" framework which accords with the notion of transdisciplinarity (notwithstanding the recursive notion that philosophy is a discipline – although some disciplines such as philosophy and education can be identified as having considerable transdisciplinary character). Elsewhere, IT regards itself as *postdisciplinary* in that it can be used in conjunction with "transdisciplinarity" or any discipline. That may be so; however, transdisciplinaries are also used in conjunction with disciplines; and, as with all transdisciplinarities, IT provides an overarching orientation.
[31] The term "transdisciplinarity" was first used by Piaget, Morin, and Jantsch in the early 1970s (Hampson 2010). Integrative scholar, Julie Thompson Klein (2004), defines it as "a common system of axioms for a set of disciplines" (p.515). She notes that it can include comprehensive paradigms such as Marxism, broad interdisciplinary fields such as cultural studies, and even synoptic disciplines such as philosophy.
[32] Noting that apt "interference" includes such manoeuvres as indicated by post-positivism.
[33] The meta-approaches have not to date fulfilled their potential in this regard.
[34] Of course, apt understanding of differences between the approaches would form part of this possibility.
[35] Though not comprehensive
[36] From an integrative perspective, context is a form of content
[37] For example, in the current context: What pertinent questions might be asked? Could the above list benefit from modification? Given, Arlin's (1975) address of "problem-finding" as a postformal manoeuvre, such questions should remain present throughout research engagement (as continuously generative but without becoming debilitating through "ongoing revision fatigue").
[38] There are, of course, a host of (other) specific topics, including many explicitly arising from the three meta-approaches discussed here. Unfortunately, space does not permit further discussion.
[39] "One turn" of knowledge: the poetics of complexity-within-unity